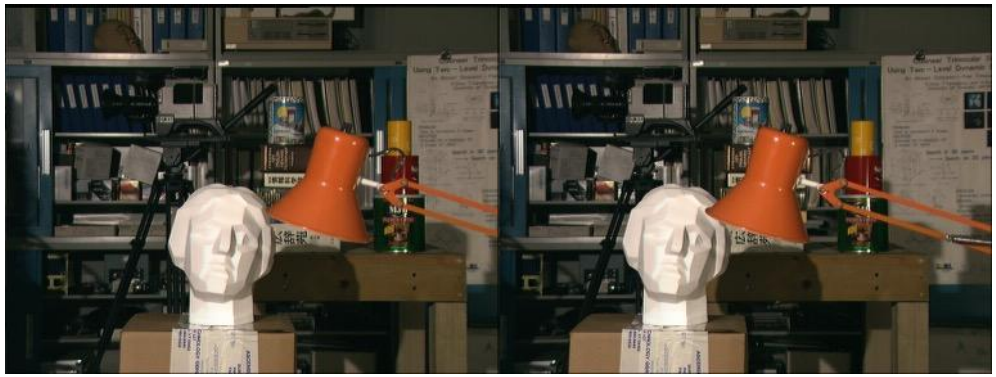
CS5187: Vision and Image

Multi-view stereo

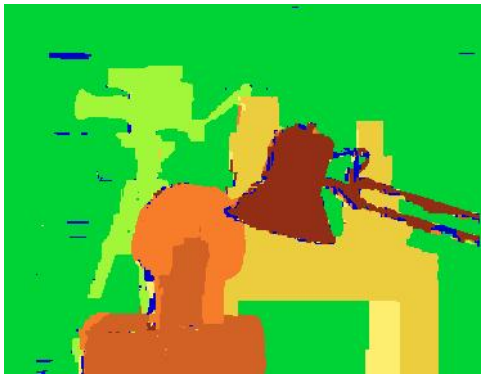


Multi-Camera Array

Before: Binocular (Two-View) Stereo



Left-right (rectified) stereo pair



Computed disparity map

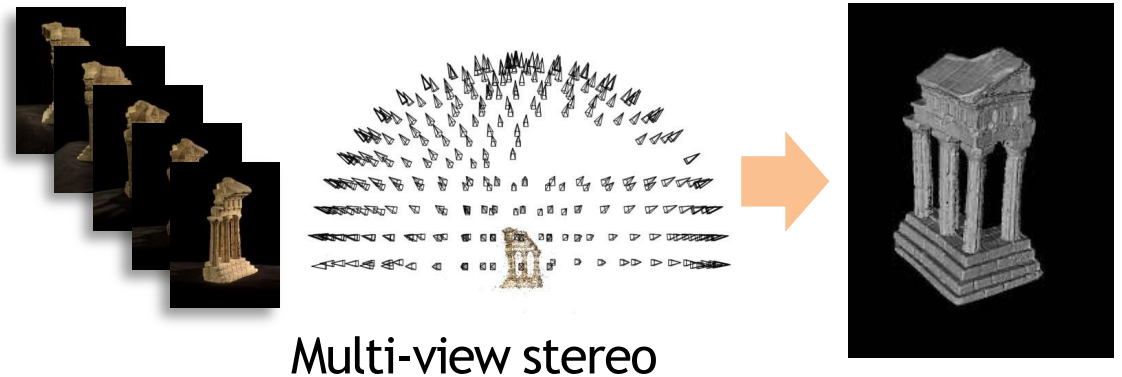
Useful for robot perception and navigation, video effects, etc.

Now: Multi-view Stereo

Problem formulation: given several images of the same object or scene, compute a representation of its 3D shape



Binocular Stereo



Multi-view stereo

Multi-view Stereo



Point Grey's Bumblebee XB3



Point Grey's ProFusion 25

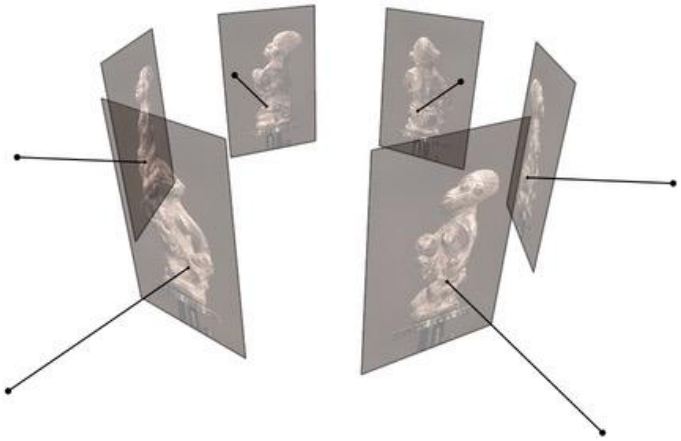


CMU's Panoptic Studio

Multi-view Stereo

Input: calibrated images from several viewpoints (known intrinsics and extrinsics / projection matrices)

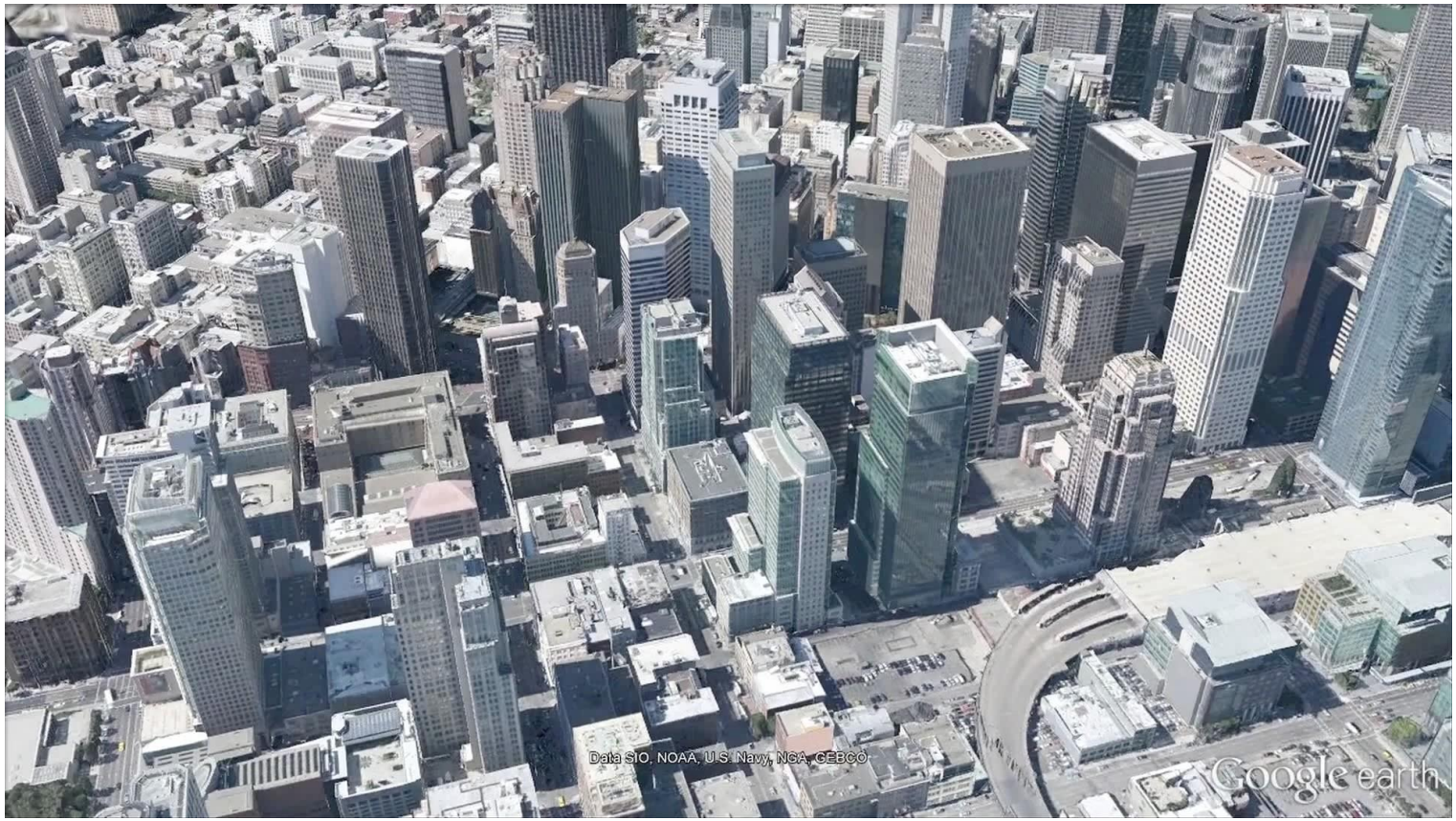
Output: 3D object model



Figures by Carlos Hernandez

We'll talk more about how to calibrate multiple cameras soon

Applications


Data SIO, NOAA, U.S. Navy, NGA, GEBCO
Google earth

Whistle in the Form of Female Figure *600 AD - 900 AD*

Details Los Angeles County Museum of Art



Los Angeles County Museum of Art | Sculpture | Mexico

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